Long-range Transport Methods for PM

Part I: Concepts and Trajectory Statistical Methods

Student reading material accompanying the presentation

This reading material covers Part I only: the conceptual basis of long-range transport analysis, back trajectories, trajectory statistical methods, CPF/PSCF, CWT, practical mapping choices, and interpretation rules. It does not cover the Part II software installation or R practical workflow.

Learning objectives

- Explain why measured particulate matter (PM) at a receptor site may include local, regional, and long-range contributions.
- Describe what back trajectories represent and why they should be interpreted statistically rather than individually.
- Explain the workflow of trajectory statistical methods (TSM).
- Interpret CPF/PSCF and CWT maps correctly, including their main strengths and limitations.
- Identify the practical settings that should be reported when producing trajectory-based maps.

01. Course roadmap

Part I introduces the conceptual path from measured PM concentrations to trajectory-based interpretation. The overall goal is to understand how air-mass history can be combined with measured pollutant concentrations to identify transport pathways and potential source regions. The main methods emphasized are CPF/PSCF and CWT, because both connect receptor concentrations with the residence time of air masses over geographic grid cells.

The logic of the course is simple: first, understand why transported PM matters; second, understand what trajectory endpoints represent; third, learn how these endpoints are joined with measured concentrations; and finally, interpret statistical maps cautiously and reproducibly.

Key idea: trajectory statistical methods do not replace chemical analysis or source apportionment. They add a meteorological transport layer that helps interpret where air masses came from before high or low concentration events.

02. Why study long-range transport?

Particulate matter measured at a monitoring station is rarely produced by only one type of source. It can include emissions from nearby traffic, heating, or industry; particles and precursor gases formed regionally; and material transported over hundreds or thousands of kilometres. For example, a receptor site in southern Europe may be affected by local traffic emissions on some days and by African dust or regional secondary aerosol formation on others.

This matters for air-quality interpretation. If transported aerosols or precursor gases are important, a decrease in local emissions may not immediately produce a proportional decrease in measured PM. Long-range transport analysis helps distinguish between local control problems and episodes that are partly driven by regional or transboundary atmospheric processes.

- Local contributions can include road traffic, residential heating, construction, resuspension, and nearby industry.
- Regional contributions often include secondary sulfate, nitrate, ammonium, and secondary organic aerosol formed during transport.
- Natural transport can include desert dust, sea salt, wildfire smoke, volcanic aerosol, or biological particles.

03. Air-mass history: the core idea

A back trajectory estimates the pathway followed by an air parcel before it arrived at a receptor site. The calculation starts at the receptor, at a specified date, hour, location, and arrival height, and then runs backward in time using gridded meteorological fields. The resulting trajectory is represented as a sequence of endpoints with latitude, longitude, height, and time information.

Back trajectories are useful because they connect a measurement time to an estimated air-mass history. However, they are not exact paths. They depend on meteorological data resolution, interpolation, model assumptions, arrival height, and the duration selected for the calculation. For this reason, a single trajectory should be treated as contextual information rather than proof of transport from a specific source region.

Best practice: do not base conclusions on one trajectory. Use many trajectories and analyze the statistical patterns that emerge.

04. What does a trajectory model need?

Trajectory models such as HYSPLIT require a set of user-defined inputs. The receptor latitude and longitude define the location where the air mass arrives. The arrival time should match the air-quality measurement time, usually in UTC. The starting height determines the vertical level of the arriving air parcel, and the run duration determines how far backward the trajectory is calculated. The model also needs a meteorological dataset, such as GDAS or GFS, to describe the evolving wind fields.

The model output is typically a trajectory endpoint table. For trajectory statistical methods, this endpoint table is joined with the pollutant concentration measured at the arrival time. In other words, every endpoint along one trajectory inherits the concentration measured when that trajectory arrived at the receptor.

Input / output	Role in the analysis
Receptor location	Latitude and longitude of the monitoring site.
Arrival time	UTC date and hour matching the measurement or sample period.
Starting height	Arrival height, usually expressed as metres above ground level or sea level.
Run duration	Backward time period, commonly 72-120 h for regional PM analysis.
Meteorology	Gridded meteorological data used to calculate the path.
Endpoint table	Output containing date, date2, hour increment, latitude, longitude, height, and sometimes pressure.

05. Why trajectory statistical methods?

A single back trajectory can show the estimated path of one arriving air mass, but it is not sufficient to identify a source region. Trajectories are uncertain, and their paths can change when a different arrival height,

meteorological dataset, or run duration is selected. A single path also cannot separate transport influence from local emissions occurring at the receptor.

Trajectory statistical methods reduce this limitation by using large ensembles of trajectories. Instead of asking where one air parcel came from, TSM asks whether many endpoints repeatedly occupy certain geographic regions before high concentrations, or whether air passing over certain grid cells is statistically associated with higher measured concentrations. This ensemble approach makes the analysis more robust and more suitable for environmental interpretation.

- Single trajectories are useful for episode context but are sensitive to model and meteorological choices.
- Large trajectory ensembles reveal recurring transport pathways and potential source-region patterns.
- TSM results should be interpreted as statistical associations, not as direct proof of emission from a specific point.

06. Trajectory statistical methods: workflow

A typical TSM workflow combines pollutant measurements and meteorological transport calculations. First, PM mass or chemical species are measured at the receptor site. Second, back trajectories are calculated for the corresponding arrival times. Third, trajectory endpoints are joined with the measured concentration at the arrival time. Fourth, endpoints are assigned to geographic grid cells. Finally, statistics such as CPF/PSCF or CWT are calculated for each grid cell.

1. Measure PM or chemical species at the receptor site.
2. Calculate back trajectories for the corresponding sample or measurement times.
3. Join each trajectory with the pollutant concentration measured at its arrival time.
4. Divide the geographic domain into grid cells and count trajectory residence in each cell.
5. Calculate a statistic such as CPF/PSCF or CWT and map the result.

Output maps identify regions and pathways statistically associated with elevated receptor concentrations. They do not, by themselves, identify exact emission sources.

07. Method family: what each map answers

Different trajectory-based outputs answer different questions. A trajectory frequency map answers where air masses passed most often. Cluster analysis groups similar air-mass pathways and helps characterize dominant meteorological regimes. CPF/PSCF maps focus on high-concentration events and ask where air masses passed before those events. CWT maps use concentration-weighted averaging and ask which grid cells are associated with higher average receptor concentrations.

Map / method	Main question
Frequency map	Where did air masses pass most often?
Cluster analysis	Which air-mass pathways are similar?
CPF / PSCF	Where did trajectories pass during high concentration events?
CWT	Which grid cells are associated with higher average receptor concentrations?

Because each method emphasizes a different aspect of the data, they should be used together rather than interpreted in isolation. Agreement between multiple views gives more confidence than a single map alone.

08. CPF / PSCF: the probability view

CPF/PSCF is a probability-based trajectory statistic. It asks: when concentrations at the receptor are high, where did the air masses pass? For each grid cell, the method counts all trajectory endpoints that pass through the cell and then counts the subset of those endpoints associated with high-concentration arrivals at the receptor.

A high CPF/PSCF value means that a grid cell is frequently crossed by air masses before high concentration days or hours. This makes the method useful for identifying transport pathways and geographic regions linked to high pollution episodes. However, the result is a conditional probability, not a concentration value. It should not be read as the amount of PM emitted from that cell.

In openair trajectory analysis, CPF/PSCF-style mapping is implemented using trajLevel(..., statistic = "pscf").

09. CPF / PSCF formula and interpretation

CPF/PSCF is often expressed as a ratio between high-event residence and total residence in each grid cell:

$$\mathbf{CPF_{ij} / PSCF_{ij} = m_{ij} / n_{ij}}$$

Here, n_{ij} is the total number of trajectory endpoints in grid cell (i,j), and m_{ij} is the number of endpoints in that same cell for trajectories associated with high receptor concentrations. The definition of “high” is usually controlled by a percentile threshold, commonly the 90th percentile, although this should be tested and reported.

- Use CPF/PSCF when the question is focused on high-concentration episodes.
- Report the concentration threshold or percentile used to define “high”.
- Apply minimum-bin or weighting criteria so that cells with very few endpoints do not dominate the interpretation.
- Remember that a high value shows conditional probability, not emission strength.

10. CWT: the concentration-weighted view

CWT, or concentration weighted trajectory analysis, asks a different question from CPF/PSCF. Instead of only counting high events, CWT uses the measured pollutant concentration associated with each trajectory. Each trajectory endpoint carries the concentration measured when that trajectory arrived at the receptor. Grid cells that are crossed by trajectories linked to high receptor concentrations receive higher CWT values.

This makes CWT useful for identifying regions where air-mass residence is associated with higher average receptor concentrations. In contrast to CPF/PSCF, CWT preserves concentration intensity and does not rely only on a high-event threshold. It is therefore a useful complement to CPF/PSCF, especially when the goal is to compare relative importance of potential source regions or transport corridors.

In openair, CWT maps are commonly produced with trajLevel(..., statistic = "cwt").

11. CWT formula and interpretation

A simplified form of the CWT statistic is:

$$\mathbf{CWT_{ij} = \Sigma(C_k\, \tau_{ijk}) / \Sigma(\tau_{ijk})}$$

C_k is the pollutant concentration measured at the arrival of trajectory k, and τ_{ijk} is the residence time of trajectory k in grid cell (i,j). The statistic therefore represents a residence-time-weighted average concentration associated with air masses passing over that cell.

A high CWT value means that air parcels passing over that grid cell are, on average, associated with higher receptor concentrations. This is helpful for ranking potential source regions or transport corridors, but it still requires external interpretation. High values can be influenced by outliers, grid size, the density of trajectories, smoothing choices, and local events at the receptor that are not actually caused by long-range transport.

12. CPF/PSCF versus CWT

CPF/PSCF and CWT are related but not interchangeable. CPF/PSCF focuses on the frequency of air-mass passage during high events, while CWT estimates the average concentration associated with air passing over each grid cell. Because they answer different questions, they often provide complementary evidence.

Feature	CPF / PSCF	CWT
Core question	Where were trajectories during high events?	Which cells are linked to higher average concentrations?
Output	Probability / conditional probability	Concentration-weighted field
Strength	Highlights high-event pathways	Preserves concentration intensity
Main caution	Threshold choice matters and sparse cells can mislead	Hot cells can reflect outliers or trajectory-frequency effects
Teaching shortcut	How often during high events?	How high on average?

13. Practical choices before mapping

The settings selected before producing CPF/PSCF or CWT maps can substantially affect the result. These choices should be made deliberately, tested where possible, and reported transparently. The appropriate settings depend on the pollutant, the region, the receptor location, the temporal resolution of the data, and the scientific question.

- Trajectory length: 72-120 h is common for regional PM analysis, but shorter or longer periods may be justified depending on the pollutant and region.
- Starting height: test near-surface and elevated arrival heights, because transport pathways can differ vertically.
- Grid size: choose a grid coarse enough to contain sufficient endpoints but fine enough to show regional structure.
- High threshold: for CPF/PSCF, the 90th percentile is common, but sensitivity to threshold should be checked.
- Minimum bin / weighting: sparse grid cells should be excluded or down-weighted to avoid unstable values.
- Smoothing: smoothing improves visualization but can blur sharp gradients and should not be used to hide sparse data.

Always report: meteorological dataset, receptor coordinates, arrival height, trajectory duration, grid increments, pollutant, percentile threshold, minimum-bin criteria, and smoothing choices.

14. Interpretation rules

Trajectory maps are powerful tools, but they can be misinterpreted. A high CPF/PSCF or CWT value should be read as a statistical association between receptor concentrations and air-mass residence over a geographic region. It should not be presented as direct proof that emissions from that grid cell caused the observed concentration. The best interpretations combine trajectory maps with chemical tracers, source apportionment, emission inventories, satellite observations, local meteorology, and known event information.

Do	Do not
Use many trajectories.	Interpret one trajectory as proof.
Compare CPF/PSCF and CWT.	Ignore local sources at the receptor.
Check chemical tracers and source-apportionment evidence.	Over-read cells with very few endpoints.
Test seasons, heights, trajectory length and grid size.	Forget trajectory uncertainty.
Cross-check with emissions, dust/wildfire events and meteorology.	Hide sensitivity choices or uncertain assumptions.

Core interpretation principle: trajectory maps suggest potential source regions and transport pathways; they do not prove source emissions alone.

summary

- Long-range transport can substantially influence PM and PM chemical components at a receptor site.
- Back trajectories estimate air-mass history but are uncertain and should be interpreted statistically.
- Trajectory statistical methods combine many trajectory endpoints with measured pollutant concentrations.
- CPF/PSCF highlights regions crossed before high-concentration events.
- CWT highlights regions associated with higher average receptor concentrations.
- Both methods require sensitivity testing and should be interpreted together with chemistry and meteorology.

Review questions

6. Why can local PM concentrations remain high even when local emissions decrease?
7. Why is a single back trajectory not enough to identify a source region?
8. What information must be specified before calculating a back trajectory?
9. What does a high CPF/PSCF value mean? What does it not mean?
10. How is CWT different from CPF/PSCF?
11. Which practical settings should be reported when publishing a trajectory-statistical map?
12. Name three types of external evidence that can help validate the interpretation of a CWT or CPF/PSCF map.

Key terms

Term	Meaning
Receptor site	The monitoring location where PM is measured.
Back trajectory	An estimated backward path of an air parcel before arrival at the receptor.
Endpoint	A point along a trajectory with latitude, longitude, height and time.

Residence time	The amount of trajectory time spent in a grid cell.
CPF / PSCF	A probability-based statistic showing where trajectories passed during high concentration events.
CWT	A concentration-weighted statistic showing which cells are associated with higher average receptor concentrations.
Minimum bin	A threshold used to reduce the influence of grid cells with too few endpoints.
Sensitivity analysis	Testing how results change when settings such as trajectory height, duration, grid size or threshold are modified.